Three-level hierarchical POMDP: an executed test of the N-level template

The 2-level hierarchical POMDP (sec. 17.9) couples location inference to a single global context.

Three-level hierarchical... (part 2)

The N-level architecture uses `build_nlevel_world` from the `fedference.pomdp` module and provides a parameterized stack of levels; the canonical 3-level example couples location (L1; 9 states) to a context variable (L2;

Three-level hierarchical... (part 3)

2 states: `quiet` / `alert`) and further to a meta-context variable (L3; 2 states: `low_threat` / `high_threat`) that gates the L2 prior.

Three-level hierarchical... (part 4)

$$\tilde{D}_{\mathsf{L2}} = \sum_{k} q_{\mathsf{L3}}[k]\, p_{\mathsf{L2|L3}}[k] \tag{38}$$

Three-level hierarchical... (part 5)

$$\tilde{D}_{\mathsf{L1}} = \sum_{c} q_{\mathsf{L2}}[c]\, p_{\mathsf{L1|L2}}[c] \tag{39}$$

Three-level hierarchical... (part 6)

The inference algorithm (`fedference.pomdp.nlevel_infer`) performs 4 passes of top-down / bottom-up alternating minimization: the top-down pass propagates empirical priors from L3 $\rightarrow$ L2 $\rightarrow$ L1 via eq. 38 and eq. 39;

Three-level hierarchical... (part 7)

the bottom-up pass updates each level's belief from the marginal evidence contributed by the level below.

We compare two conditions over 960 seeded trials with 4 agents at sensor acuity 0.85.

Three-level hierarchical... (part 8)

Flat condition. Agents ignore all hierarchy and infer location under a uniform prior.

Three-level condition. Agents run 4 alternating-minimization iterations across all three levels before federating.

Three-level hierarchical... (part 9)

The measured location accuracies are 0.982 (flat) and 0.969 (3-level), a gap of -0.014.

Three-level hierarchical... (part 10)

Across 128 independent seeds the 3-level location accuracy is 0.977 (SD 0.005; 95 % CI 0.976–0.977) versus flat 0.983 (95 % CI 0.982–0.984), a mean accuracy gap of -0.007 (95 % CI -0.007—0.006; Wilcoxon signed-rank $p = 4.95 \times 10^{-19}$, effect size $r = 0.918$, large).

Three-level hierarchical... (part 11)

The signed mean gap and its interval describe the location-accuracy cost of this configured three-level condition relative to the flat baseline.

The 3-level condition additionally reports context accuracy 0.697 and meta-context accuracy 0.547.

Three-level hierarchical... (part 12)

Against the two-state chance baseline of 0.5, location is recovered and the intermediate context latent is resolved well above chance, but the meta-context latent is only marginally above chance — the weakest of the three levels — and is therefore *not* convincingly recovered here.

Three-level hierarchical... (part 13)

The study thus demonstrates that the generic N-level alternating-minimization runs and federates end-to-end and recovers the fastest (location) and intermediate (context) latents; full recovery of the slowest (meta-context) level is left open.

Three-level hierarchical... (part 14)

The full figure comparing 2-level and 3-level belief dynamics is fig. 31. The declarative layer specification used by the generic constructor is documented in the supplement (sec. 17.12). For the effect of acuity and colony size on these results, see sec. 17.13.